ASSIGNMENT – 4

Execute the following tasks, and then prepare a report as instructed.

Task 1. Basic Familiarisation with Wireshark:

a. Open Wireshark and familiarise yourself with the interface.

b. Identify the different panes (Packet List, Packet Details, Packet Bytes).

c. Locate the available interfaces to capture on.

Task 2: Basic Packet Capture:

a. Start capturing packets on your primary interface.

b. Visit a webpage containing multiple objects.

c. Stop the capture after a few seconds.

d. Analyse the captured packets and identify the different protocols used (e.g., TCP, UDP, HTTP, DNS).

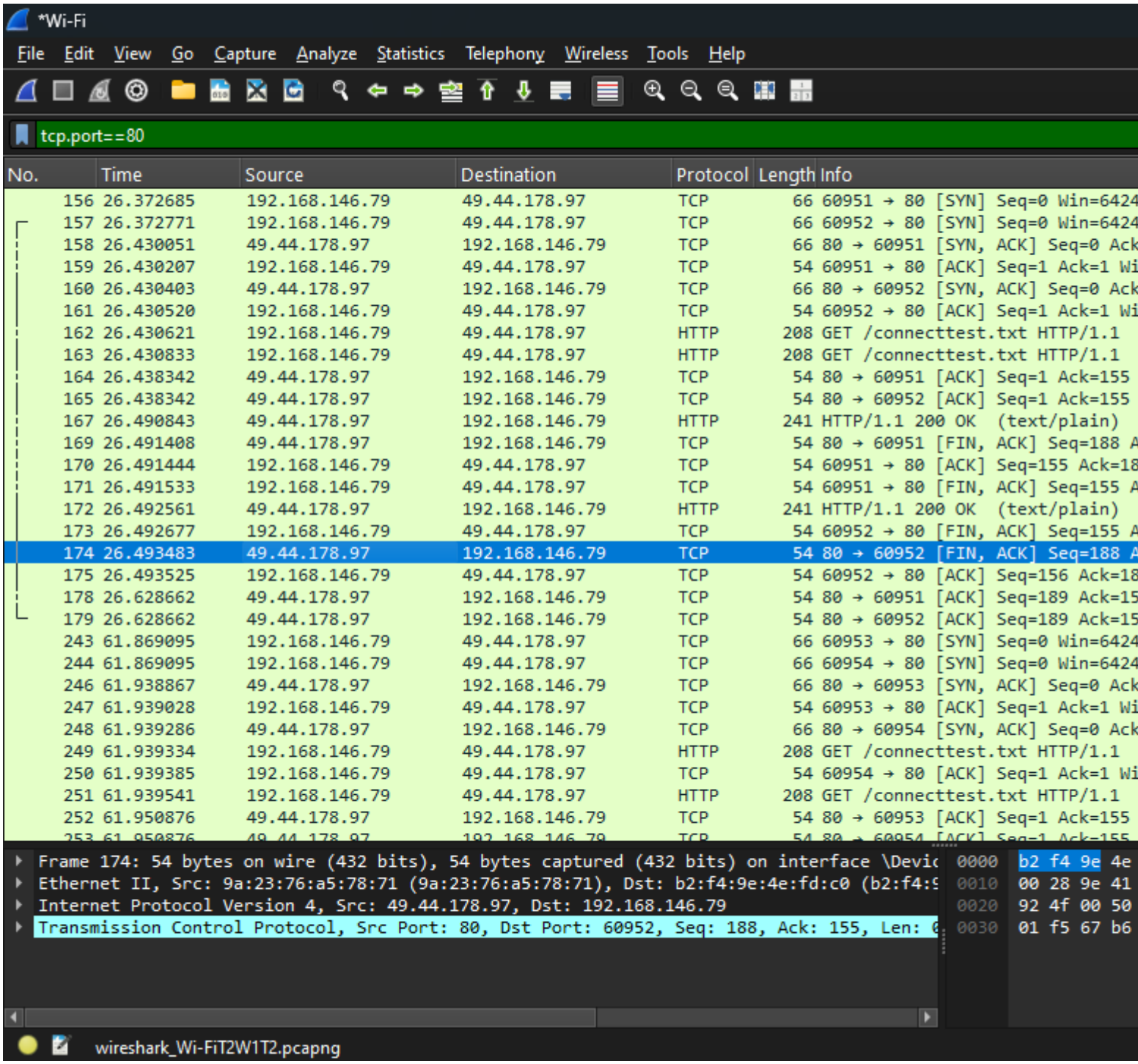
Task 3 : Filtering Packets:

a. Use Wireshark's display filter to view only HTTP packets.

b. Use the filter to view only DNS packets.

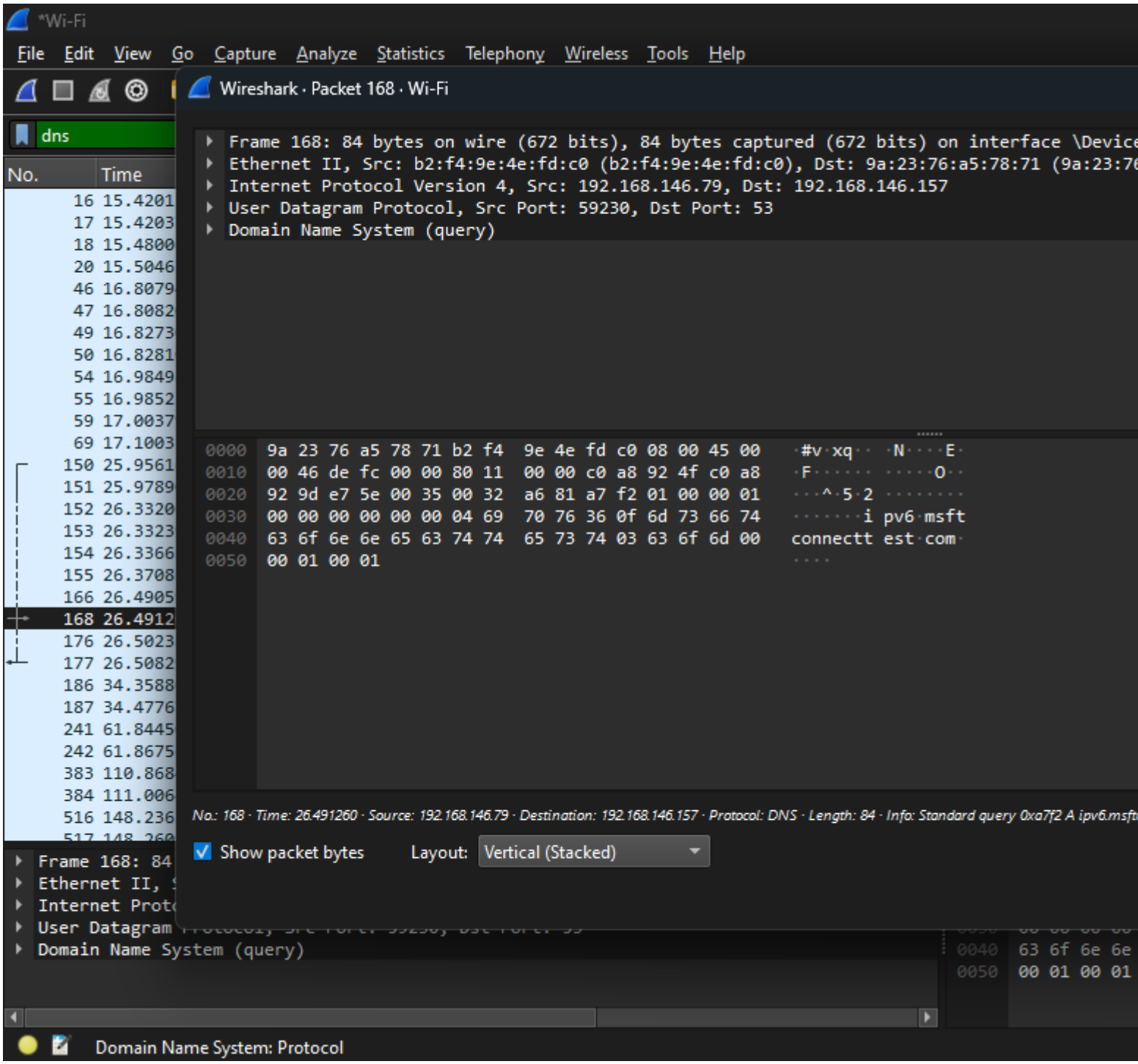
Now prepare a report with answers of the following, along with relevant screenshots and annotations (wherever required). A single snapshot may contain multiple annotations:

1. Use a filter to view TCP packets going to or coming from port 80.



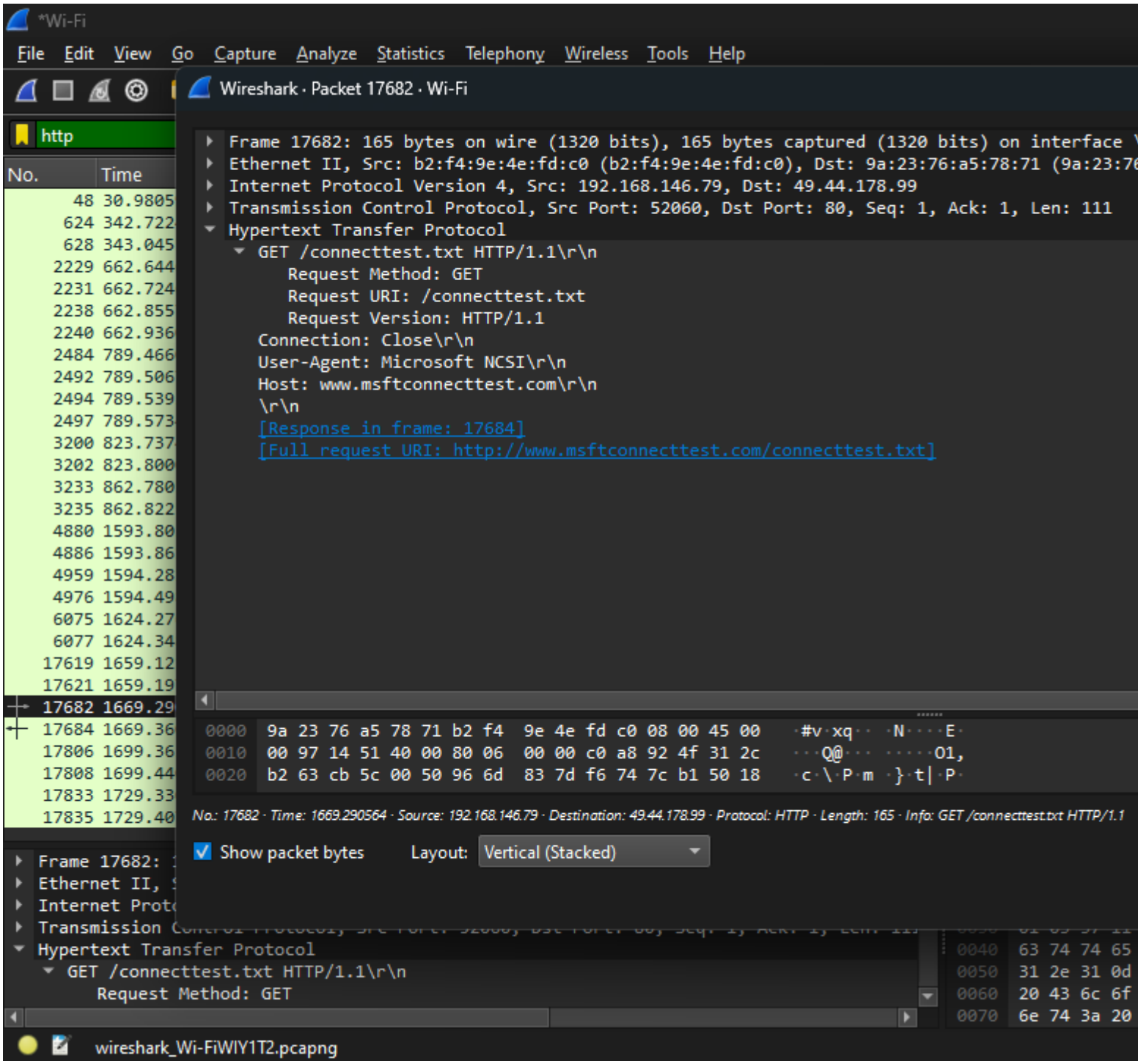
This screenshot shows that there are a total of 40 TCP present which is coming from or going to port 80. The filter used is – tcp.port == 80

2. Select a DNS packet and analyse its details. Identify the domain name being queried.



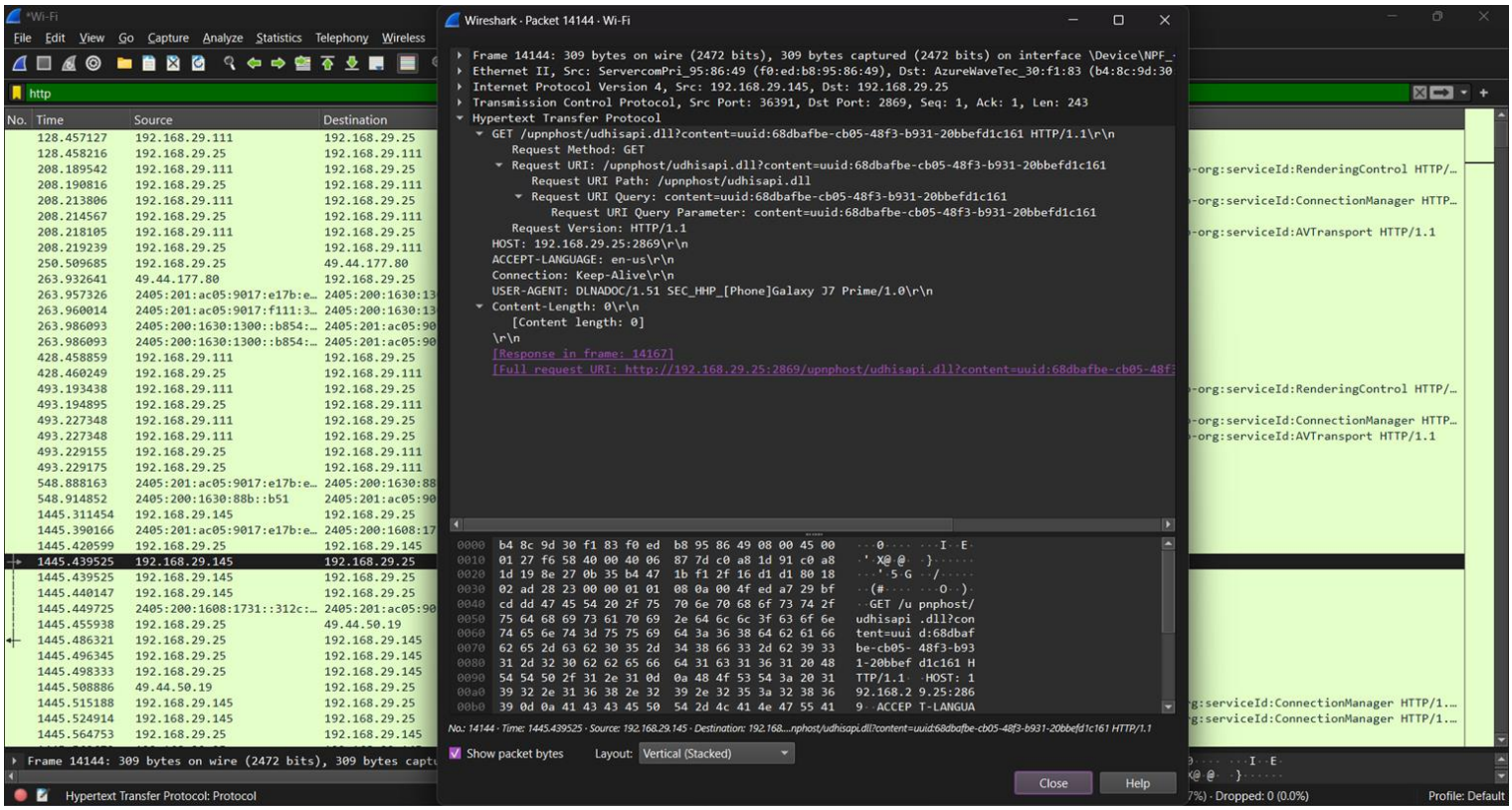
This screenshot shows the domain name being queried for the given selected DNS as seen in the screenshot with the source address 192.168.146.79 and serial No. 168

3. Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?



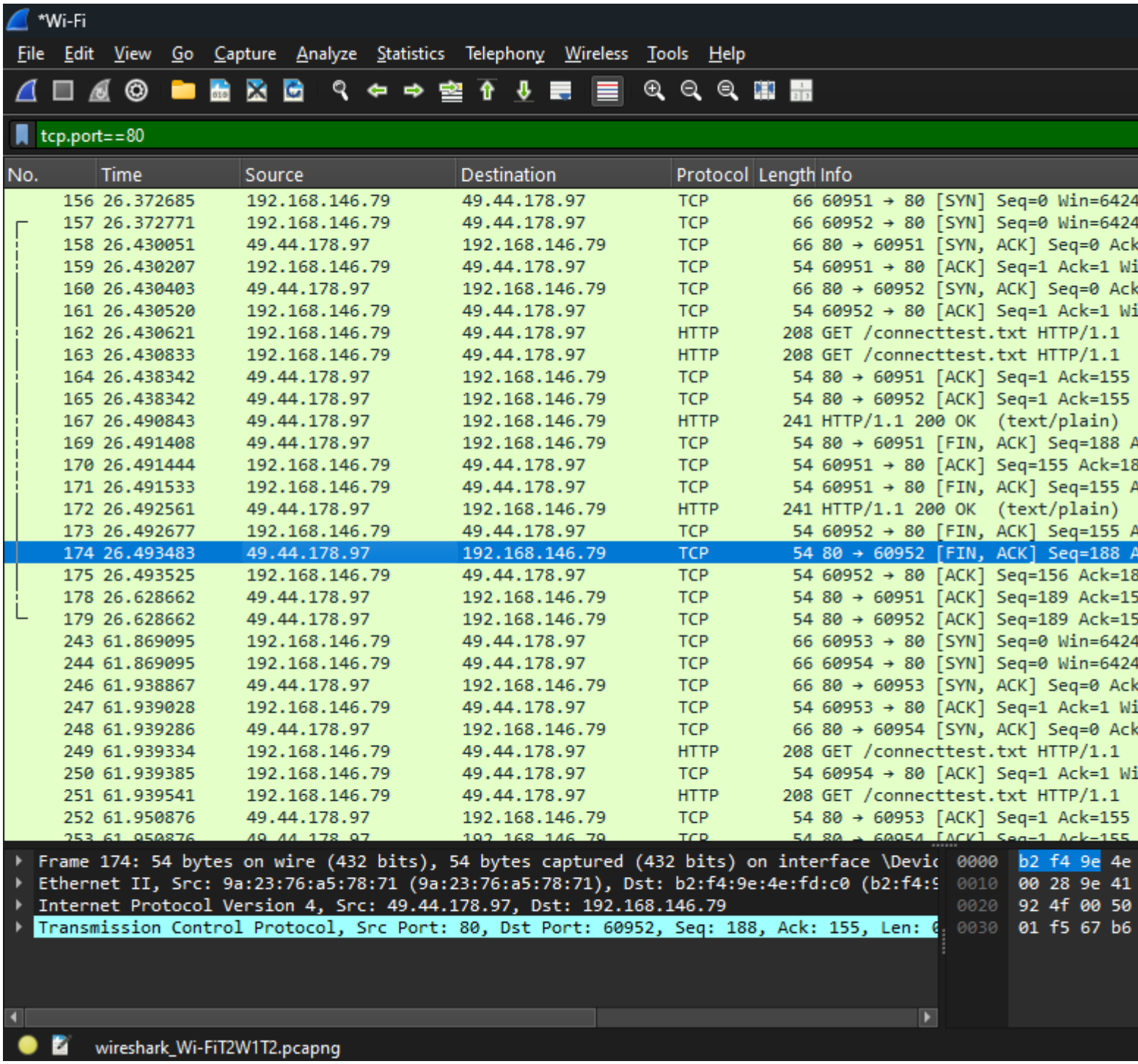
The HTTP version in use is 1.1

4. What languages (if any) does your browser indicate that it can accept to the server?



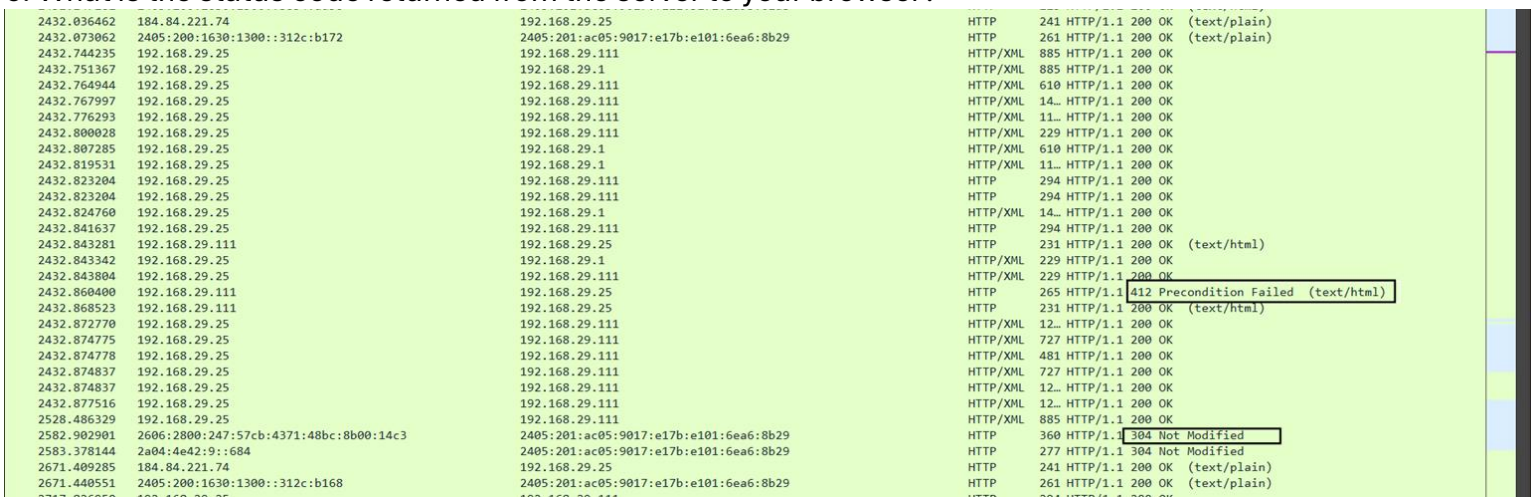
The `Accept-Language` header is sent by the client (browser or application) to specify the preferred language(s) for the content it receives from the server. `en-us` indicates that the client prefers content in US English.

5. What is the IP address of your computer? Of the server you visited?



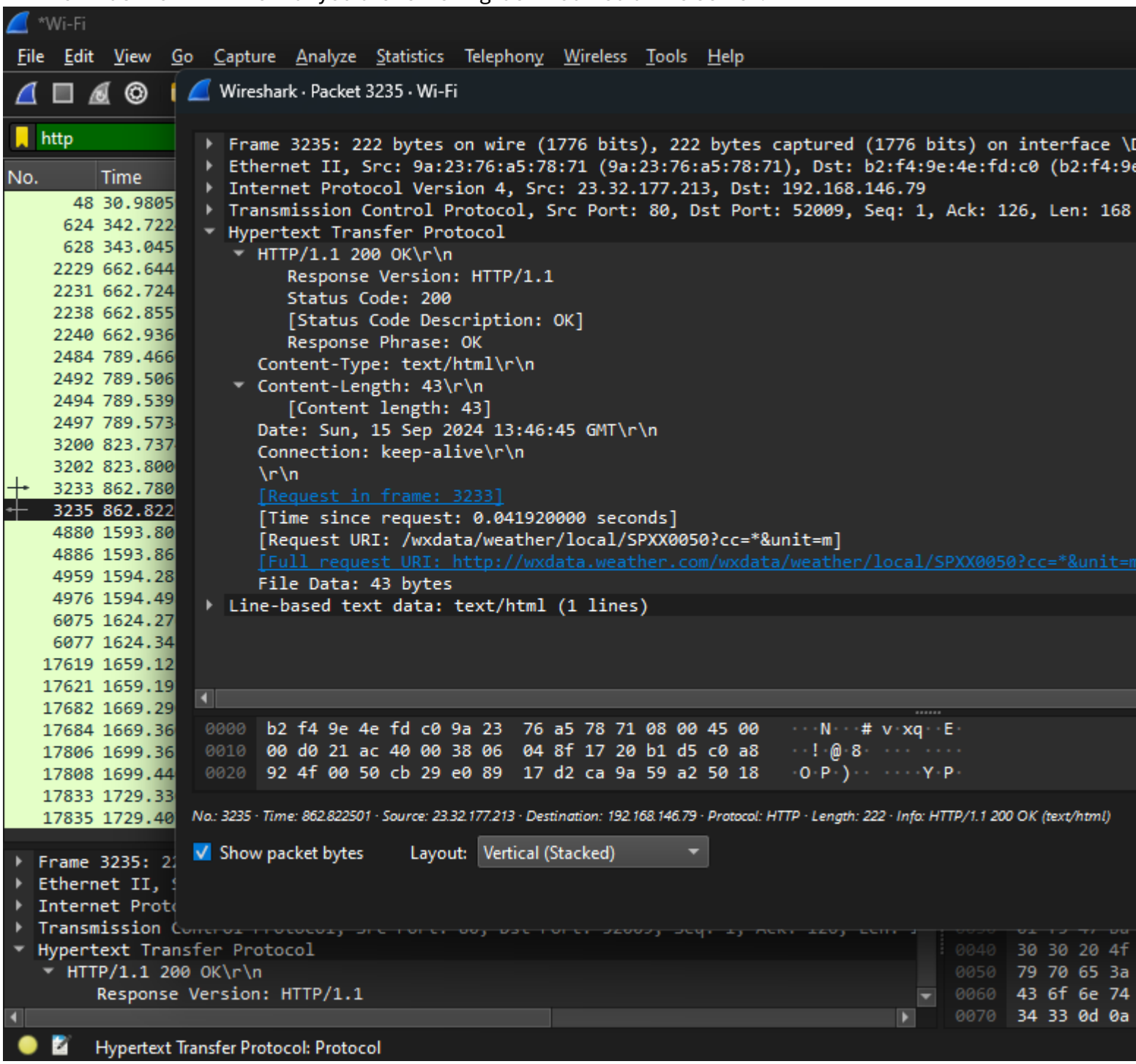
The addresses that are in the source IP addresses of the servers and the computers visited.
The addresses that are in the destination IP addresses of the servers (or the computers) visited.

6. What is the status code returned from the server to your browser?



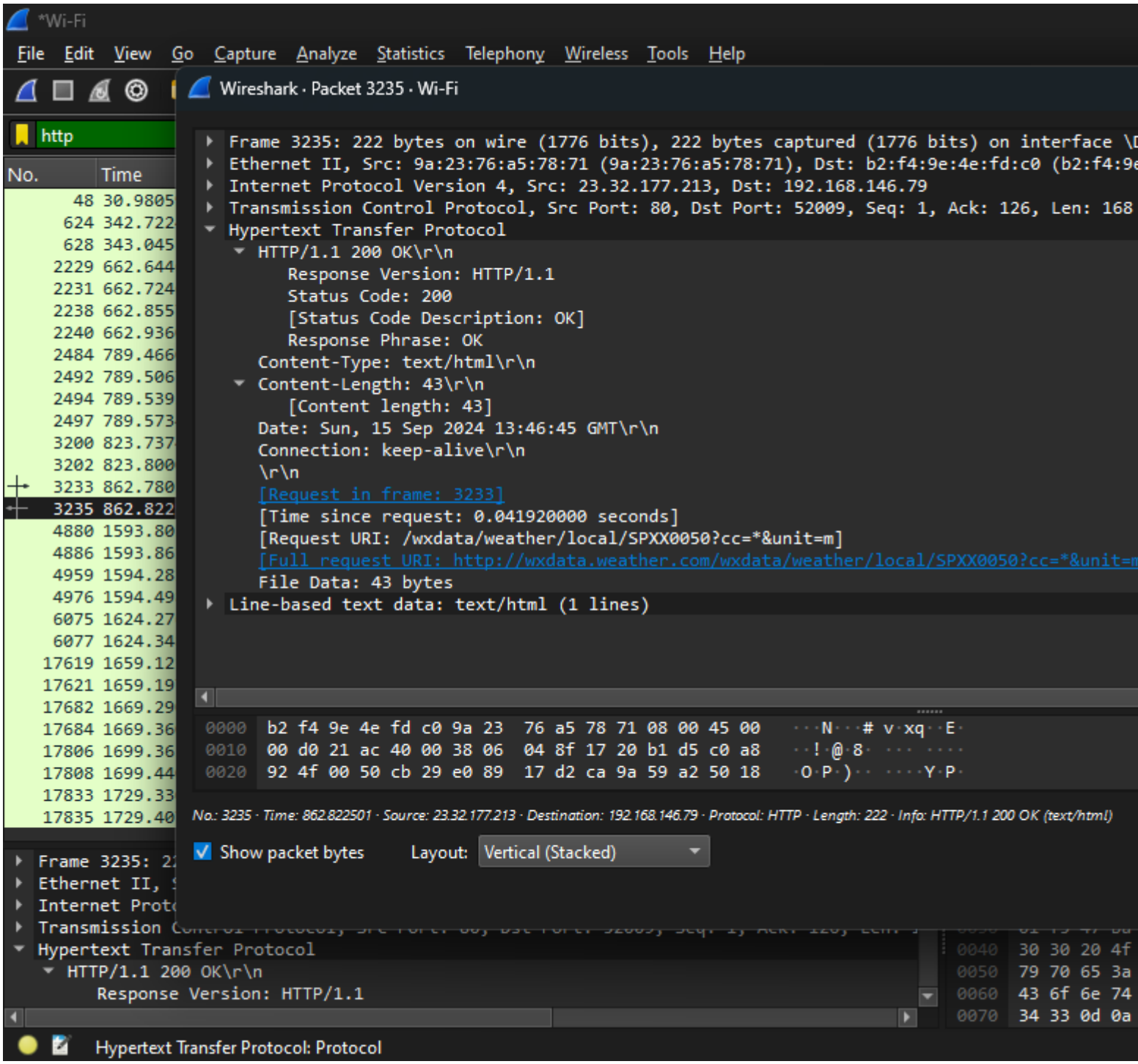
A total of 3 types of status codes can be observed from the above screenshot.
Status code – 200 OK, 412-Precondition Failed 304 and 304 Not Modified.

7. When was the HTML file that you are retrieving last modified at the server?



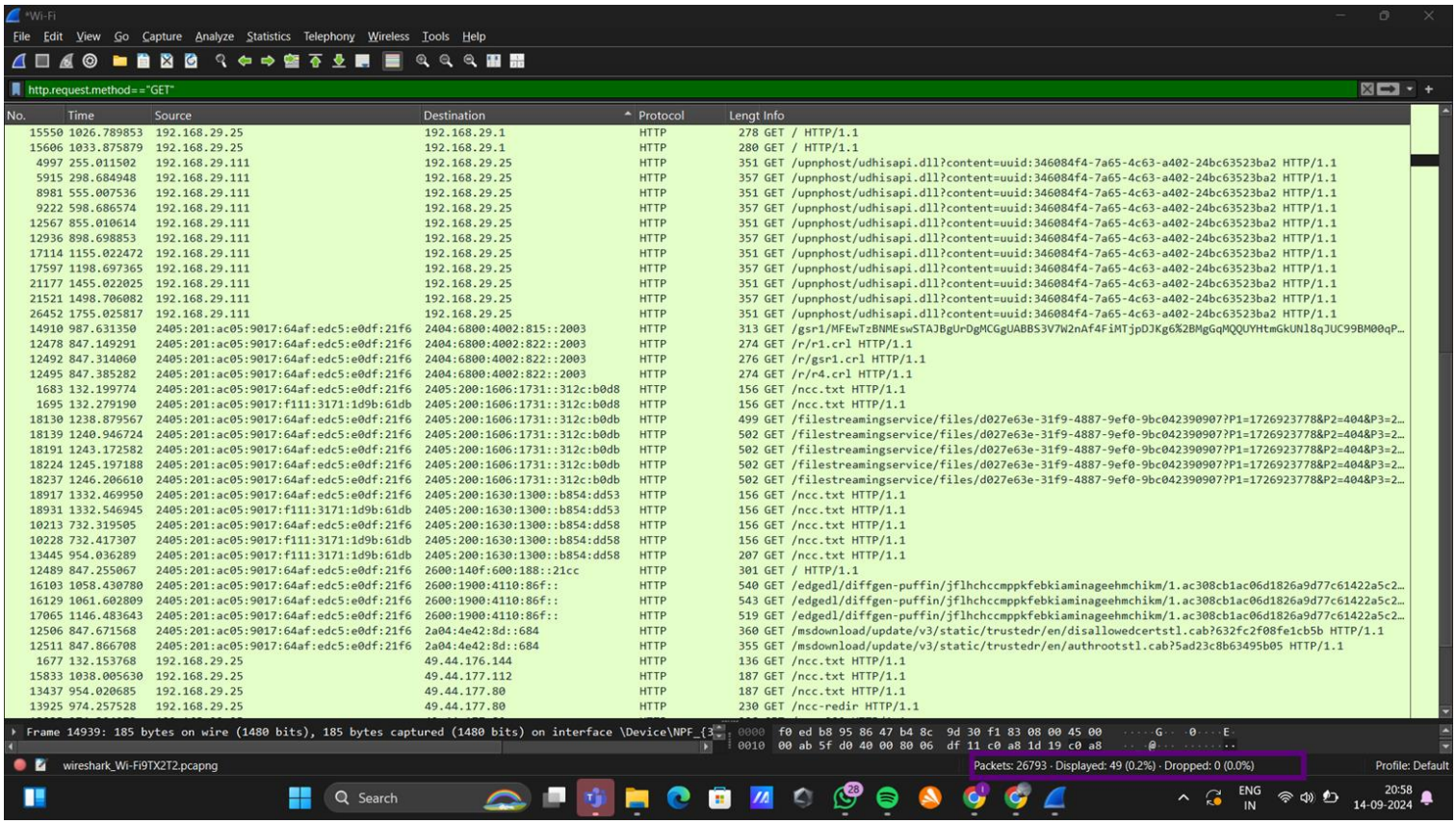
The last modified time for the html file being retrieved is Sun,15 Sep 2024 13:46:45.

8. How many bytes of content are being returned to your browser?



For the same HTTP response packet as the previous one, the content length is 43.

9. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

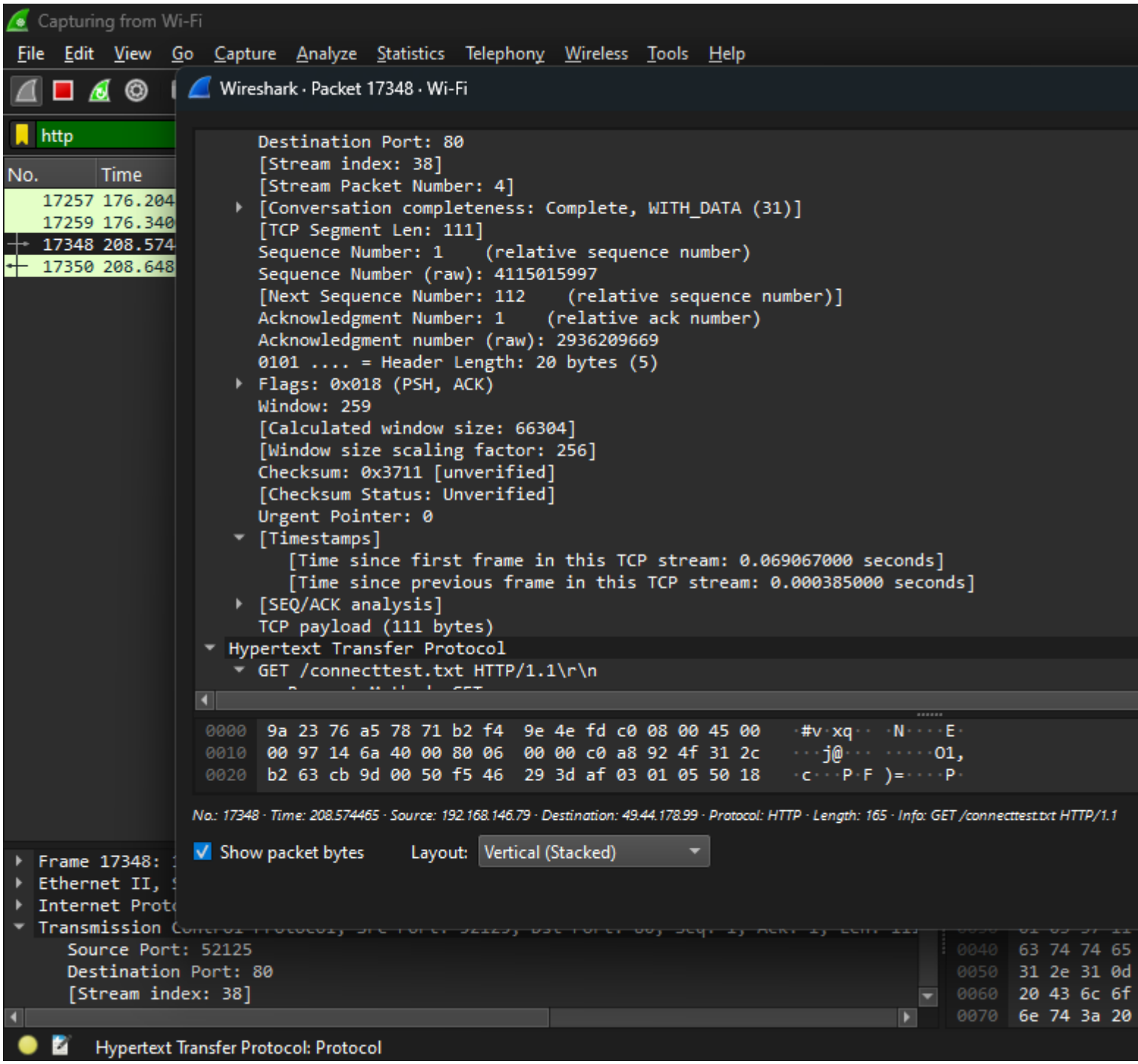


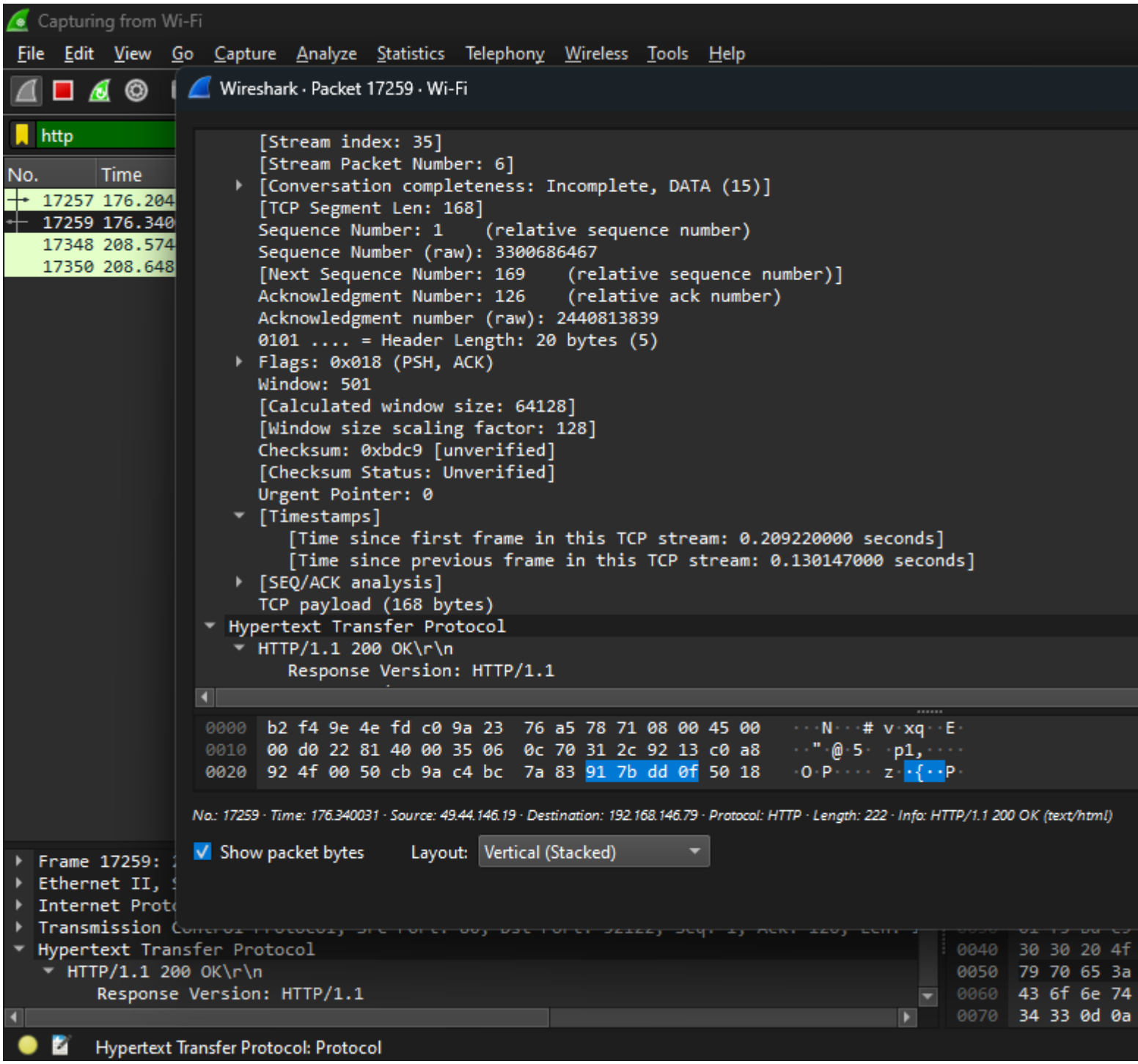
The total count of HTTP GET request messages are given in violet annotated box below in the screenshot. The Internet address to which these GET requests were sent are 192.168.29.1,198.168.29.25, 49.44.177.80,49.44.176.112 and 49.44.176.144

10. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

Below there are two packets bearing serial numbers 17259 and 17348.

The timestamps of HTTP GET request packet bearing serial number 17259 and 17348 is given below and is thus observed.


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Wireshark · Packet 17348 · Wi-Fi
http
No. Time
17257 176.204
17259 176.340
17348 208.574
17350 208.648
Destination Port: 80
[Stream index: 38]
[Stream Packet Number: 4]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 111]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4115015997
[Next Sequence Number: 112 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2936209669
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 259
[Calculated window size: 66304]
[Window size scaling factor: 256]
Checksum: 0x3711 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[Time since first frame in this TCP stream: 0.069067000 seconds]
[Time since previous frame in this TCP stream: 0.000385000 seconds]
[SEQ/ACK analysis]
TCP payload (111 bytes)
Hypertext Transfer Protocol
GET /connecttest.txt HTTP/1.1\r\n
0000 9a 23 76 a5 78 71 b2 f4 9e 4e fd c0 08 00 45 00
0010 00 97 14 6a 40 00 80 06 00 00 c0 a8 92 4f 31 2c
0020 b2 63 cb 9d 00 50 f5 46 29 3d af 03 01 05 50 18
No.: 17348 · Time: 208.574465 · Source: 192.168.146.79 · Destination: 49.44.178.99 · Protocol: HTTP · Length: 165 · Info: GET /connecttest.txt HTTP/1.1
Show packet bytes
Layout: Vertical (Stacked)
Frame 17348:
Ethernet II,
Internet Prot
Transmission
Source Port: 52125
Destination Port: 80
[Stream index: 38]
0040 63 74 74 65
0050 31 2e 31 0d
0060 20 43 6c 6f
0070 6e 74 3a 20
Hypertext Transfer Protocol: Protocol



On close observation, we can thus conclude that the two images got downloaded serially.